Cambridge Assessment International Education

Cambridge IGCSE™

BIOLOGY **9700/53**

Paper 5 Planning, Analysis and Evaluation **October/November 2022**

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
underline	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore

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Question	Answer	Marks
1(a)(i)	to allow, oxygen / air, to enter for respiration **or** because, bacterium / *B. subtilis,* respires aerobically / AW **;**	**1**
1(a)(ii)	as, population growth of bacteria / bacterial growth, increases / AW, the, cloudier / more turbid, the culture broth **or** measuring the, turbidity / cloudiness, by eye against a cross in background / with a colorimeter / turbidity meter / colour chart **;**	**1**
1(b)	replace antibiotic with, (distilled) water / nutrient solution **;**	**1**
1(c)	*can improve:* add, catalase / peroxidase / manganese(IV) oxide **;** *explain why:* results / effect / killing, *B. subtilis* / bacteria , is only due to, honey / MGO **or** results / effect / killing, *B. subtilis* / bacteria, is not due to hydrogen peroxide **;**	**2**
1(d)	*description or table or diagram for 10%:* 1 Correct method to dilute stock (honey) solution to 10% **;** *table* 2 At least five dilutions including 10% all with correct volumes of stock and nutrient solution including units **;**	**2**
1(e)	*independent variable* MGO / Manuka / honey, concentration **;** *dependent variable* turbidity **;**	**2**

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Question	Answer	Marks
1(f)	*predict:* lower concentration, tubes / AW, will be, turbid / cloudy **or** higher concentration, tubes / AW, will be, clear / not cloudy **or** the lower the concentration the more, turbid / cloudy, the tubes / AW **; ora** *explain:* MGO / (Manuka) honey, inhibits the growth of / kills, *B. subtilis / bacteria* **or** lower concentrations have less, ability / MGO, to, inhibit the growth of / kill, *B. subtilis / bacteria* **; ora**	**2**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(g)</td><td>
any five from:

1 same / stated, volume of, culture / broth used ;

2 same / stated, volume of Manuka honey ;

3 mix / stir, honey (concentrations) and, culture / broth (prior to starting) ;

4 method to measure turbidity ;

5 idea that: write down / note / record, which concentration(s) the broth, is clear / is clearest ;

any two from 6–8 ;;
6 method of maintaining temperature (10°C – 50°C if stated)

7 same / stated, time (left in incubator) (12–48 hours if stated)

8 use a buffer to maintain pH (pH 4–10 if stated)

9 repeat at least twice / 3 replicates and finding mean ;

10 named hazard and risk and precaution ;

<table>
<tr><th>Hazard</th><th>Risk</th><th>Precaution</th></tr>
<tr><td>Live cultures / bacterium / B. subtilis</td><td>Allergy / Infection / irritation / toxic</td><td>Wear gloves / goggles / mask / PPE</td></tr>
<tr><td>Honey</td><td>Allergy</td><td>Wear gloves / goggles / mask / PPE</td></tr>
<tr><td>Broth</td><td>Irritant / allergy</td><td>Wear gloves / goggles / mask / PPE</td></tr>
</table>
</td><td>5</td></tr>
<tr><td>1(h)(i)</td><td>there is no difference in the effect / effectiveness, of Manuka (honey) and Germania (honey) ;</td><td>1</td></tr>
</table>

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Question	Answer	Marks
1(h)(ii)	yes yes **;** no	**1**
1(h)(iii)	*results:* Manuka honey, kills more bacteria / is more effective (than Germania) **; ora** statistical analysis: first three / 1, 2 and 3, are significantly different **or** last two / 4 and 5, not significantly different against MRSA **or** 1 is only significant at p = 0.05 **or** 2 / 3, are significant at the, p = 0.01 / p = 0.001 **;**	**2**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)</td><td>any one from:

To ensure no gossypol in any of the cows / to remove gossypol from cows
or
idea of: feeding on same diet standardises cows (for start of experiment) ;</td><td>1</td></tr>
<tr><td>2(b)(i)</td><td>any three from:

1 the (mean) milk yields are, very close / similar, between groups ;

2 no statistical test
or
should carry out a, statistical test / t-test ;

3 overlap of SE (then no significant difference) ;

4 detail of which groups, are not / may be, significantly different ;

<table>
<tr><th>not significantly different</th><th>may be significantly different</th></tr>
<tr><td>A and B</td><td>A and E</td></tr>
<tr><td>A and C</td><td>B and E</td></tr>
<tr><td>A and D</td><td>C and E</td></tr>
<tr><td>B and C</td><td>D and E</td></tr>
<tr><td>B and D</td><td></td></tr>
<tr><td>C and D</td><td></td></tr>
</table>
</td><td>3</td></tr>
</table>

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Question	Answer	Marks
2(b)(ii)	*any **two** from:* *enough evidence because:* 1 trend that as gossypol (in diet) increases, milk yield increases **; ora** 2 paired data quote (to support) **;** *not enough evidence because:* 3 *idea that:* no information that differences in milk yield are caused by (differences in) gossypol **;** 4 *idea that:* no information on, nutritional content / AW, of , no gossypol diet / GM cottonseed **;** 5 *idea that:* there is no cottonseed in A so cannot compare with (diets) B to E **or** There is only soybean in A so cannot compare with (diets) B to E **;** 6 no statistical test **or** should carry out a, statistical test / *t*-test **;**	**2**
2(c)(i)	*there is a relationship because:* as free (gossypol) intake increases, plasma (gossypol) concentration increases **or** shows a positive correlation **;**	**1**

Question	Answer	Marks
2(c)(ii)	*allow description of diet instead of group* **B**, etc. *any **three** from:* 1 (Group) E has the highest risk (of toxicity) **or** (Group) C has the lowest risk (of toxicity) **;** 2 sequence of risk, highest to lowest, is **E →B →D →C ;** 3 *ref. to* higher free gossypol **and** (–) isomer values for highest risks **or** *ref. to* lower free gossypol **and** (–) isomer values for lower risks **;** 4 paired data quote from Table 2.2 to support mp 2 or 3 **;** 5 free gossypol will be easily absorbed and more toxic (than bound gossypol) **or** Bound gossypol is less easily absorbed and less toxic / not toxic **;** 6 *idea that:* uncertain how much free gossypol turns into bound gossypol when eaten **;** 7 AVP **;**	**3**